1) Simple Batch distillation separates 0.6kmol of a binary feed that is 70.0mol % methanol & 30.0mol % water, final still pot is 10.0mol methanol

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Quiz #3.

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$$\boxed{\text{Mass bal: } F = W + D}$$

$$\boxed{\text{comp bal } x_F F = x_W W + x_{Davg} D}$$

$$x_{Davg} = \frac{x_F F - x_W W_{final}}{D_{TOTAL}}$$

Simpsons Rule for area need $\frac{1}{y-x}$

methanol $0.7 \to 0.1$

x (methanol liquid)	y	1/y−x
0.1	0.418	3.14465
0.4	0.729	3.03951
0.7	0.870	5.88235

$n = 2$ $\Delta x = 0.7 - 0.1$

$$\text{Simpsons} = \frac{(0.7-0.1)}{2 \cdot 3}\left[3.14465 + 4(3.03951) + 5.88235\right]$$

area = 2.11851

$W_{final} = F \exp(-\text{area}) = (0.6\text{kmol})(\exp(-\text{area})) =$

$D_{TOT} = F - W_{final}$

$$\boxed{W_{final} = 0.0721 \text{ kmol}}$$

$$\boxed{D_{TOTAL} = 0.5279 \text{ kmol}}$$

$$\boxed{x_{Davg} = 0.782}$$

↑ from above

$$x_{Davg} = \frac{(0.6)(0.7)^{\leftarrow \text{kmol} \leftarrow \text{initial}} - (0.1)(0.0721)^{\leftarrow \text{final}}}{0.5279}$$